Exercise 1C

1. Reduce the following fractions to their simplest forms:

(i) $\frac{2}{8}$ (ii) $\frac{7}{21}$ (iii) $\frac{16}{24}$ (iv) $\frac{14}{35}$ (v) $\frac{24}{60}$ (vi) $\frac{42}{54}$

2. Express each of the following fractions as a mixed number:

(i) $\frac{6}{5}$ (ii) $\frac{8}{3}$ (iii) $\frac{11}{4}$ (iv) $\frac{16}{3}$ (v) $\frac{29}{3}$ (vi) $\frac{43}{8}$

3. Express each of these mixed numbers as a fraction:

(i) $1\frac{2}{3}$ (ii) $2\frac{2}{5}$ (iii) $3\frac{3}{4}$ (iv) $5\frac{2}{9}$ (v) $6\frac{2}{3}$ (vi) $3\frac{7}{10}$

4. Copy and complete the following:

(i) $\frac{2}{5} = \frac{4}{\square}$ (ii) $\frac{3}{4} = \frac{9}{\square}$ (iii) $\frac{4}{7} = \frac{\square}{21}$

(iv) $\frac{5}{8} = \frac{15}{\square}$ (v) $\frac{5}{6} = \frac{\square}{18}$ (vi) $\frac{3}{5} = \frac{\square}{25}$

5. Simplify each of the following:

(i) $\frac{3}{7} + \frac{2}{7}$ (ii) $\frac{1}{9} + \frac{3}{9} + \frac{4}{9}$ (iii) $\frac{9}{10} + \frac{7}{10} - \frac{3}{10}$

6. Express each of the following as a single fraction:

(i) $\frac{3}{4} + \frac{1}{2}$ (ii) $\frac{1}{3} + \frac{1}{6}$ (iii) $\frac{1}{2} + \frac{1}{3}$ (iv) $\frac{1}{2} + \frac{7}{8}$

(v) $\frac{3}{4} + \frac{2}{5}$ (vi) $\frac{2}{3} - \frac{1}{4}$ (vii) $\frac{5}{9} - \frac{1}{6}$ (viii) $\frac{11}{12} - \frac{1}{18}$

7. Simplify the following:

(i) $\frac{1}{2} + 3\frac{1}{4}$ (ii) $1\frac{3}{8} + 2\frac{3}{4}$ (iii) $1\frac{1}{6} + 2\frac{1}{3}$ (iv) $3\frac{1}{5} + 1\frac{3}{10}$

(v) $3\frac{3}{4} - 1\frac{1}{2}$ (vi) $2\frac{3}{4} - 1\frac{2}{3}$ (vii) $2\frac{3}{4} - 1\frac{7}{8}$ (viii) $3\frac{3}{4} - 2\frac{1}{3}$

8. Multiply the following fractions:

(i) $\frac{4}{5} \times \frac{1}{4}$ (ii) $\frac{4}{5} \times \frac{1}{2}$ (iii) $\frac{18}{25} \times \frac{5}{6}$ (iv) $\frac{7}{8} \times \frac{4}{5}$

(v) $1\frac{2}{3} \times \frac{4}{5}$ (vi) $3\frac{3}{4} \times \frac{2}{3}$ (vii) $4\frac{1}{2} \times 1\frac{5}{6}$ (viii) $1\frac{3}{4} \times 2\frac{2}{7}$

9. Divide the following fractions:

(i) $\frac{3}{4} \div \frac{1}{3}$ (ii) $\frac{5}{9} \div \frac{2}{3}$ (iii) $\frac{4}{5} \div \frac{1}{10}$ (iv) $\frac{8}{21} \div \frac{4}{7}$

(v) $\frac{9}{14} \div \frac{3}{7}$ (vi) $2\frac{1}{2} \div \frac{1}{4}$ (vii) $3\frac{3}{4} \div \frac{2}{3}$ (viii) $3\frac{1}{8} \div 3\frac{3}{4}$

10. Simplify each of the following:

(i) $\left(\frac{2}{3} - \frac{1}{2}\right) \div \frac{2}{3}$ (ii) $\left(3\frac{1}{3}\right) \times \left(2\frac{1}{4} - \frac{3}{5}\right)$ (iii) $\left(2 - \frac{2}{3}\right) \div \frac{8}{9}$

11. You decide to dye your hair various colours for Halowe'en. If you dye $\frac{1}{4}$ of it green, $\frac{2}{5}$ of it blue and $\frac{1}{5}$ of it black, how much of your original hair colour is left?

12. You walk $\frac{4}{5}$ km to school and $\frac{4}{5}$ km back each day.
How many kilometres do you walk to and from school each week?

13. Jordan walks $\frac{1}{2}$ km to school. Paula walks $\frac{7}{8}$ km to school.
How much further does Paula walk than Jordan?

14. $\frac{1}{4}$ of a playground is a basketball court, $\frac{3}{8}$ is used as a soccer pitch.
How much is left for other activities?

15. Mia swims a race in $35\frac{7}{10}$ seconds. Orla swims the race in $34\frac{3}{10}$ seconds.
How much faster is Orla?

16. A bucket contains $4\frac{3}{4}$ litres of water. If $2\frac{3}{8}$ litres spill, how much water is left in the bucket?

17. A 20 cent coin is $1\frac{3}{10}$ cm wide. Placed end to end, how long would 5 coins be?